

Truly Innovative TECH

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

Radiation-tolerant Ethernet transceiver and MCU

Ethernet has become common in spacecraft for enabling high-speed data communication between satellites and spacecraft while reliably performing in varied radiation levels. Supporting this



is the VSC8541RT radiation-tolerant Ethernet transceiver solution along with the SAM3X8ERT radiation-tolerant microcontroller that includes Arm Cortex-M3 core processor and embedded Ethernet controller. Both the devices are COTS-based, allowing designers to significantly reduce development time and cost for use in satellite platforms, payloads for data and sensor bus control, remote terminal communication, space vehicle networks, and module connectivity in space stations.

Microchip Technology
www.microchip.com

Field programmable wireless power solution

Claiming to deliver a highly-integrated 60W wireless power in a single-chip



transmitter/receiver, the P9418 is suited for portable computing devices such as smartphones, laptops, and notebooks. The wireless solution can transmit or receive an AC power signal through magnetic induction while also offering advanced telemetry and charging protocols required for high-power applications. Notable features include embedded 32-bit ARM Cortex-M0 processor, support for I2C 400kHz standard interface, encrypted bi-directional communications, and WPC 1.2.4 compliance.

Renesas Electronics
www.renesas.com

Automotive capacitive sensor for hands-on detection on steering wheel

For enhanced safety of assisted driving systems, automotive manufacturers are gradually shifting towards a capacitive



sensing technique, which incorporates separation of resistive and capacitive parts of the impedance measured over the sensor. This is a significant improvement from the charge-discharge method of capacitive sensing. Making this possible is the new AS8579 capacitive sensor that reliably performs hands-on detection on the steering wheel—even when the driver wears gloves. It also prevents generation of false detection signals due to the presence of moisture or humidity.

ams
www.ams.com

Digital coronavirus testing device for rapid results

Thanks to improved software, the VivaLytic analysis device, a rapid coronavirus testing tool that uses polymerase chain reactions (PCR), can now detect



a positive sample and deliver faster test results in less than thirty minutes—right where the sample is collected. The VivaLytic has a sensitivity of 98 percent and a specificity of 100 percent. The new update is available in a simple and straightforward manner for the SARS-CoV-2 through Vivasuite cloud platform, meeting strict security standards and data privacy at all times.

Bosch
www.bosch.com

Secured wireless charging IC

While wireless charging is convenient, an inaccurate power supply can be harmful to the handheld's battery lifetime and, in the worst case, the user as well. Therefore,

